

ORDER MANAGEMENT

StyleNova – Fashion Products E-Commerce Order Management System

1.1 Orders Table

Attribute	Data Type	Constraint
Order_ID	INT	Primary Key
Customer_ID	INT	Foreign Key
Order_Date	DATE	NOT NULL
Total_Amount	NUMBER(10,2)	NOT NULL

1.2 Order_Details Table

Attribute	Data Type	Constraint
Order_Detail_ID	INT	Primary Key
Order_ID	INT	Foreign Key
Product_ID	INT	Foreign Key
Quantity	INT	NOT NULL
Unit_Price	NUMBER(10,2)	NOT NULL
Subtotal	NUMBER(10,2)	NOT NULL

2. Table Relationships

The Orders table stores the main order information of StyleNova customers. The Order_Details table stores the products included in each order. One customer can place many orders, and one order can contain many order-detail records. Order_ID connects Orders with Order_Details, while Product_ID connects Order_Details with the Product table.

3. Table Creation

3.1 Create Orders Table

```
SQL> CREATE TABLE SN_Orders (  
2     Order_ID INT PRIMARY KEY,  
3     Customer_ID INT NOT NULL,  
4     Order_Date DATE NOT NULL,  
5     Total_Amount NUMBER(10,2) NOT NULL,  
6     FOREIGN KEY (Customer_ID) REFERENCES Customer(CUSTOMERID)  
7 );
```

Table created.

3.2 Create Order_Details Table

```
SQL> CREATE TABLE SN_Order_Details (  
 2     Order_Detail_ID INT PRIMARY KEY,  
 3     Order_ID INT NOT NULL,  
 4     Product_ID INT NOT NULL,  
 5     Quantity INT NOT NULL,  
 6     Unit_Price NUMBER(10,2) NOT NULL,  
 7     Subtotal NUMBER(10,2) NOT NULL,  
 8     FOREIGN KEY (Order_ID) REFERENCES SN_Orders(Order_ID),  
 9     FOREIGN KEY (Product_ID) REFERENCES Product(Product_ID)  
10 );
```

Table created.

4. Order Data Insertion

```
SQL> INSERT INTO SN_Orders  
 2 (Order_ID, Customer_ID, Order_Date, Total_Amount)  
 3 VALUES  
 4 (1001, 108, SYSDATE, 1798.00);
```

1 row created.

```
SQL>  
SQL> INSERT INTO SN_Orders  
 2 (Order_ID, Customer_ID, Order_Date, Total_Amount)  
 3 VALUES  
 4 (1002, 224, SYSDATE, 1299.00);
```

1 row created.

```
SQL>  
SQL> INSERT INTO SN_Orders  
 2 (Order_ID, Customer_ID, Order_Date, Total_Amount)  
 3 VALUES  
 4 (1003, 567, SYSDATE, 1499.00);
```

1 row created.

```
SQL>  
SQL> COMMIT;
```

Commit complete.

5. Order_Details Data Insertion

```
SQL> INSERT INTO SN_Order_Details
  2 (Order_Detail_ID, Order_ID, Product_ID, Quantity, Unit_Price, Subtotal)
  3 VALUES
  4 (1, 1001, 101, 2, 599.00, 1198.00);

1 row created.

SQL>
SQL> INSERT INTO SN_Order_Details
  2 (Order_Detail_ID, Order_ID, Product_ID, Quantity, Unit_Price, Subtotal)
  3 VALUES
  4 (2, 1001, 103, 1, 600.00, 600.00);

1 row created.

SQL>
SQL> INSERT INTO SN_Order_Details
  2 (Order_Detail_ID, Order_ID, Product_ID, Quantity, Unit_Price, Subtotal)
  3 VALUES
  4 (3, 1002, 102, 1, 1299.00, 1299.00);

1 row created.

SQL>
SQL> INSERT INTO SN_Order_Details
  2 (Order_Detail_ID, Order_ID, Product_ID, Quantity, Unit_Price, Subtotal)
  3 VALUES
  4 (4, 1003, 105, 1, 1499.00, 1499.00);

1 row created.

SQL>
SQL> COMMIT;

Commit complete.
```

6. Order Modification Operations

6.1 Modify Order Amount

```
SQL> UPDATE SN_Orders
  2 SET Total_Amount = 1898.00
  3 WHERE Order_ID = 1001;

1 row updated.
```

6.2 Modify Order Date

```
SQL> UPDATE SN_Orders
  2  SET Order_Date = SYSDATE
  3  WHERE Order_ID = 1002;

1 row updated.
```

7. Order_Details Modification

```
SQL> UPDATE SN_Order_Details
  2  SET Quantity = 2,
  3      Subtotal = Quantity * Unit_Price
  4  WHERE Order_Detail_ID = 3;

1 row updated.
```

8. Customer Order History Report

```
SQL> SELECT
  2      o.Order_ID,
  3      o.Customer_ID,
  4      o.Order_Date,
  5      o.Total_Amount
  6  FROM SN_Orders o
  7  ORDER BY o.Customer_ID, o.Order_Date;
```

ORDER_ID	CUSTOMER_ID	ORDER_DAT	TOTAL_AMOUNT
1001	108	27-SEP-26	1898
1002	224	27-SEP-26	1299
1003	567	27-SEP-26	1499

```
SQL> SELECT
  2     o.Customer_ID,
  3     COUNT(o.Order_ID) AS Total_Orders,
  4     SUM(o.Total_Amount) AS Total_Order_Value
  5 FROM SN_Orders o
  6 GROUP BY o.Customer_ID
  7 ORDER BY o.Customer_ID;
```

CUSTOMER_ID	TOTAL_ORDERS	TOTAL_ORDER_VALUE
108	1	1898
224	1	1299
567	1	1499

```
SQL> SELECT
2      o.Order_ID,
3      o.Customer_ID,
4      p.Product_Name,
5      d.Quantity,
6      d.Unit_Price,
7      d.Subtotal
8  FROM SN_Orders o
9  JOIN SN_Order_Details d
10 ON o.Order_ID = d.Order_ID
11 JOIN Product p
12 ON d.Product_ID = p.Product_ID
13 ORDER BY o.Order_ID;
```

ORDER_ID	CUSTOMER_ID	PRODUCT_NAME	QUANTITY	UNIT_PRICE	SUBTOTAL
1001		Floral Dress	2	599	1198
1001		Slim Fit Jeans	1	600	600